

7.4.18

EE25BTECH11011 - Navya

Question:

If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π square units, find the equation of the circle.

$$1) x^2 + y^2 + 2x - 2y - 47 = 0$$

$$2) x^2 + y^2 + 2x - 2y - 62 = 0$$

$$3) x^2 + y^2 - 2x + 2y - 62 = 0$$

$$4) x^2 + y^2 - 2x + 2y - 47 = 0$$

Solution:

Given two lines are,

$$3x - 4y - 7 = 0, \quad (1)$$

$$2x - 3y - 5 = 0 \quad (2)$$

Since the lines are diameters, the center lies on both:

$$\mathbf{n}_1^T \mathbf{c} = d_1, \quad (3)$$

$$\mathbf{n}_2^T \mathbf{c} = d_2, \quad (4)$$

where

$$\mathbf{n}_1 = \begin{bmatrix} 3 \\ -4 \end{bmatrix}, \quad d_1 = 7, \quad \mathbf{n}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad d_2 = 5 \quad (5)$$

In matrix form,

$$\begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} \mathbf{c} = \begin{bmatrix} 7 \\ 5 \end{bmatrix} \quad (6)$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 3 & -4 & 7 \\ 2 & -3 & 5 \end{array} \right] \quad (7)$$

Perform row operations to reduce it to RREF:

$$\left[\begin{array}{cc|c} 3 & -4 & 7 \\ 2 & -3 & 5 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & -\frac{4}{3} & \frac{7}{3} \\ 2 & -3 & 5 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & -\frac{4}{3} & \frac{7}{3} \\ 0 & -\frac{1}{3} & \frac{1}{3} \end{array} \right] \quad (8)$$

$$\xrightarrow{R_2 \rightarrow -3R_2} \left[\begin{array}{cc|c} 1 & -\frac{4}{3} & \frac{7}{3} \\ 0 & 1 & -1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 + \frac{4}{3}R_2} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array} \right] \quad (9)$$

Hence, the centre is

$$\mathbf{c} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (10)$$

The general conic form of a circle is

$$g(\mathbf{x}) = \mathbf{x}^\top V \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad (11)$$

where $V = I$ for a circle. The centre satisfies

$$V\mathbf{c} + \mathbf{u} = 0 \implies \mathbf{u} = -\mathbf{c} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad (12)$$

Given the area 49π , the radius squared is

$$r^2 = 49. \quad (13)$$

Using the vector form relation for a circle:

$$r^2 = \mathbf{u}^\top \mathbf{u} - f = 2 - f \implies f = -47. \quad (14)$$

Hence, the equation of the circle in vector form is

$$\mathbf{x}^\top V \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad (15)$$

which in coordinates becomes

$$x^2 + y^2 + 2(-1)x + 2(1)y - 47 = 0, \quad (16)$$

$$\boxed{x^2 + y^2 - 2x + 2y - 47 = 0}. \quad (17)$$

